

BERGEN FLORIDA, Norway

WMO: 013170

Lat: 60.383N

Lon: 5.333E

Elev: 36

StdP: 100.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.3	-5.4	-14.8	1.0	-1.9	-12.2	1.3	-1.1	13.1	4.3	11.5	4.5	1.4	140	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	25.1	16.7	23.0	15.9	21.1	15.1	18.0	22.6	17.1	21.1	16.2	19.7	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	11.7	19.6	15.4	11.0	18.7	14.5	10.3	17.8	50.8	22.6	48.0	21.1	45.4	19.7	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.9	-8.3	28.2	2.2	2.2	-9.9	29.8	-11.3	31.1	-12.5	32.3	-14.1	33.9		
(5)			-9.0	19.3	2.2	1.3	-10.6	20.2	-12.0	21.0	-13.2	21.7	-14.8	22.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	2.3	2.3	3.8	7.3	10.6	13.4	15.9	15.7	12.9	8.8	5.1	2.9	(6)	(7)
	DBStd	5.88	3.45	3.12	2.71	3.00	3.30	2.84	2.93	2.72	2.68	3.04	3.51	3.87	(7)	(8)
	HDD10.0	1206	239	217	192	93	30	3	0	0	6	58	148	219	(8)	(9)
	HDD18.3	3634	498	450	451	332	240	151	89	91	164	297	396	477	(9)	(10)
	CDD10.0	638	0	0	0	11	49	106	182	177	92	20	2	0	(10)	(11)
	CDD18.3	28	0	0	0	0	1	3	12	10	1	0	0	0	(11)	(12)
	CDH23.3	159	0	0	0	1	12	22	79	42	3	0	0	0	(12)	(13)
	CDH26.7	25	0	0	0	0	2	1	18	4	0	0	0	0	(13)	(14)
	WSAvg	3.5	4.0	3.8	3.7	3.5	3.5	3.3	2.9	3.0	3.2	3.4	3.5	3.8	(14)	(15)
(15) Precipitation	PrecAvg	2273	211	157	155	121	104	132	144	187	270	270	272	251	(15)	(16)
	PrecMax	3104	498	336	413	219	242	269	297	368	630	557	560	568	(16)	(17)
	PrecMin	1419	37	27	16	7	18	4	18	6	65	44	76	25	(17)	(18)
	PrecStd	482	120	85	91	49	52	65	68	84	125	124	115	141	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.8	12.5	19.7	24.7	26.2	29.1	27.2	22.9	18.6	14.0	11.2	(19)	(20)
		MCWB	6.2	6.8	7.6	11.1	14.7	15.5	18.6	17.6	15.8	12.7	9.8	8.4	(20)	(21)
	2%	DB	8.5	8.3	9.9	16.6	21.4	22.9	25.5	24.3	20.2	15.8	12.1	9.8	(21)	(22)
		MCWB	6.2	6.4	6.0	9.5	12.9	14.6	17.2	17.0	13.9	11.2	9.2	7.8	(22)	(23)
	5%	DB	7.6	7.4	8.4	14.4	19.0	20.4	22.9	22.1	18.6	14.1	10.8	8.9	(23)	(24)
		MCWB	5.9	5.8	5.6	8.5	12.0	13.6	16.7	16.1	13.6	10.7	8.5	7.3	(24)	(25)
	10%	DB	6.8	6.6	7.5	12.2	16.6	18.4	20.8	20.3	17.1	13.0	9.8	8.0	(25)	(26)
		MCWB	5.3	5.2	5.2	7.6	10.7	12.9	15.6	15.4	13.1	10.2	7.7	6.5	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	8.4	8.5	12.5	15.7	16.9	19.6	19.2	17.0	13.9	11.4	9.5	(27)	(28)
		MCDB	8.6	8.9	10.7	17.4	22.6	23.0	26.5	24.1	20.5	16.6	12.6	10.3	(28)	(29)
	2%	WB	7.2	7.1	7.5	10.3	13.7	15.6	18.2	18.1	15.6	12.6	10.2	8.6	(29)	(30)
		MCDB	7.8	7.8	8.9	14.8	19.7	20.6	23.5	22.2	18.2	14.4	11.5	9.4	(30)	(31)
	5%	WB	6.4	6.3	6.6	9.2	12.4	14.5	17.2	17.2	14.6	11.7	9.1	7.6	(31)	(32)
		MCDB	7.2	7.2	7.9	12.9	17.8	19.0	21.4	20.9	17.0	13.4	10.3	8.6	(32)	(33)
	10%	WB	5.5	5.4	5.7	8.3	11.3	13.6	16.3	16.2	13.7	10.7	8.1	6.7	(33)	(34)
		MCDB	6.6	6.4	7.1	11.4	15.7	17.3	20.1	19.2	16.2	12.5	9.3	7.8	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	4.4	5.2	6.6	7.2	6.8	6.5	6.3	5.8	5.1	4.2	3.8	(35)	(36)
		MCDBR	4.1	4.4	6.5	10.4	11.3	10.5	10.4	9.1	8.4	6.6	4.6	4.2	(36)	(37)
	5% WB	MCWBR	3.5	3.5	3.9	5.1	5.1	4.6	4.3	3.9	4.0	3.8	3.6	3.7	(37)	(38)
		MCWBR	3.8	3.9	5.1	8.5	10.3	8.9	9.0	7.7	6.3	5.2	4.3	3.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.285	0.316	0.340	0.370	0.360	0.356	0.371	0.367	0.350	0.328	0.293	0.392	(39)	(40)
		taud	2.143	2.265	2.318	2.314	2.423	2.469	2.451	2.459	2.495	2.437	2.232	2.612	(40)	(41)
	Ebn at Noon		458	672	783	825	866	875	850	825	776	664	452	321	(41)	(42)
		Edh at Noon	42	67	88	106	102	99	99	91	75	57	39	29	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		0.22	0.75	1.81	3.42	4.93	5.25	4.65	3.57	2.18	1.06	0.33	0.14	(43)	(44)
	RadStd		0.04	0.13	0.28	0.35	0.47	0.63	0.44	0.35	0.27	0.16	0.05	0.02	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	-0.87	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	+0.68	N/A	N/A	N/A	N/A	-84	-123	N/A	N/A

Nomenclature: See separate page